Complex Analysis Homework 5

1. (a) Suppose $p(z) = a_n z^n + \cdots + a_1 z + a_0$ is bounded by M for $|z| = 1$. Show that each coefficient a_k is bounded by M.

(b) Suppose $p(z) = a_n z^n + \cdots + a_1 z + a_0$ is bounded by 1 for $|z| \le 1$. Show that $|p(z)| \le |z|^n$ for all z where $|z| \ge 1$.

Hint: Consider $q(z) = \dfrac{p(z)}{z^n}$ in the annulus: $1 \le |z| \le R$ for "large" R.]

Solution

(a) $|a_k| = \left| \dfrac{p^{(k)}(0)}{k!} \right| \le \dfrac{k!M}{k!} = M$

(b) Consider $q(z) = \dfrac{p(z)}{z^n}$ in the annulus: $1 \le |z| \le R$ for "large" R. Throughout $|z| = 1$, $|q(z)| = |p(z)| \le 1$ and, if $|z| = R$, then $|q(z)| = \left| a_n + \frac{a_{n-1}}{z} \cdots + \frac{a_0}{z^n} \right| \le |a_n| + \varepsilon_R \le 1 + \varepsilon_R \xrightarrow{R \to \infty} 1$.

Hence $\left| \dfrac{p(z)}{z^n} \right| = |q(z)| \le 1$ for all z where $|z| \ge 1$.

That means $|p(z)| \le |z|^n$ for all z where $|z| \ge 1$.

2. Let $P(z) = z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \quad n \ge 2,$ such that all its zeros $c_1, c_2, \cdots, c_n$ lie inside the circle $|z| = R$.

(a) Show that $\left| \dfrac{1}{P(z)} \right| \le \dfrac{1}{(R - |c_1|)(R - |c_2|) \cdots (R - |c_n|)}$

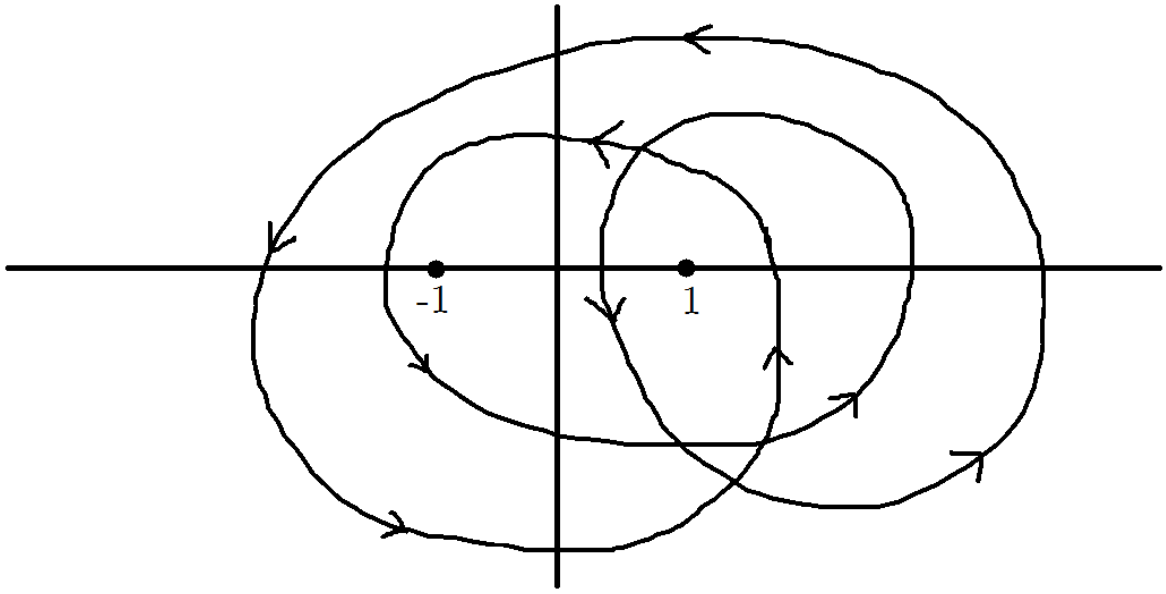
(b) Hence show that $\oint_{|z|=R} \dfrac{1}{P(z)} dz = 0.$

Solution

(a) $P(z)=(z-c_1)\cdots(z-c_n) \quad \Rightarrow \quad \left|\frac{1}{P(z)}\right| = \frac{1}{|z-c_1|\cdots|z-c_n|} \le \frac{1}{(R-|c_1|)\cdots(R-|c_n|)}$

(b) $\left|\oint_{|z|=R} \frac{1}{P(z)}dz\right| \le \frac{2\pi R}{(R-|c_1|)(R-|c_2|)\cdots(R-|c_n|)} \xrightarrow{R\to\infty} 0$. Hence $\oint_{|z|=R} \frac{1}{P(z)}dz = 0$.

3. Let C be the curve in below figure. Calculate $\int_C \frac{e^z}{z^3-z^2-z+1}dz$.



Solution

$$\int_C \frac{e^z}{z^3-z^2-z+1}dz = 3\int_{C_1} \frac{e^z}{(z-1)^2(z+1)}dz + 2\int_{C_{-1}} \frac{e^z}{(z-1)^2(z+1)}dz$$

$$= 6\pi i \frac{d}{dz}\left(\frac{e^z}{z+1}\right)\Bigg|_{z=1} + 4\pi i\left(\frac{e^z}{(z-1)^2}\right)\Bigg|_{z=-1}$$

$$= 6\pi i\left(\frac{(z+1)e^z - e^z}{(z+1)^2}\right)\Bigg|_{z=1} + 4\pi i\left(\frac{e^z}{(z-1)^2}\right)\Bigg|_{z=-1}$$

$$= 6\pi i\left(\frac{e}{4}\right) + 4\pi i\left(\frac{e^{-1}}{4}\right)$$

$$= \frac{3e+2e^{-1}}{2}\pi i$$

4. If $f(z)$ is analytic on and inside the unit circle $|z| = 1$, show that

$$(1-|z|^2) f(z) = \frac{1}{2\pi i} \oint_{|\zeta|=1} f(\zeta) \frac{1-\bar{z}\zeta}{\zeta - z} d\zeta, \quad |z| < 1,$$

and deduce that

$$(1-|z|^2) |f(z)| \le \frac{1}{2\pi} \int_0^{2\pi} |f(e^{i\theta})| d\theta, \quad |z| < 1.$$

Solution

Let $F(\zeta) = (1-\bar{z}\zeta) f(\zeta)$ where $|z| < 1$.

$$F(z) = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{F(\zeta)}{\zeta - z} d\zeta, \quad |z| < 1,$$

$$(1-|z|^2) f(z) = (1-\bar{z}z) f(z) = \frac{1}{2\pi i} \oint_{|\zeta|=1} f(\zeta) \frac{1-\bar{z}\zeta}{\zeta - z} d\zeta, \quad |z| < 1.$$

$$= \frac{1}{2\pi i} \int_0^{2\pi} f(e^{i\theta}) \frac{1-\bar{z}e^{i\theta}}{e^{i\theta} - z} ie^{i\theta} d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) \frac{1-\bar{z}e^{i\theta}}{1-ze^{-i\theta}} d\theta$$

$$\left[\because \ 1-\bar{z}e^{i\theta} = \overline{1-ze^{-i\theta}} \quad \therefore |1-\bar{z}e^{i\theta}| = |1-ze^{-i\theta}| \right]$$

$$(1-|z|^2) |f(z)| \le \frac{1}{2\pi} \int_0^{2\pi} |f(e^{i\theta})| \left| \frac{1-\bar{z}e^{i\theta}}{1-ze^{-i\theta}} \right| d\theta = \frac{1}{2\pi} \int_0^{2\pi} |f(e^{i\theta})| d\theta$$

5. Evaluate the following integrals with counterclockwise contour:

(a) $\oint_{|z|=2} \frac{z^n}{z-1} dz, \quad n \ge 0$

(b) $\oint_{|z|=1} \frac{z^n}{z-2} dz, \quad n \ge 0$

(c) $\oint_{|z|=1} \frac{\sin z}{z} dz$

(d) $\oint_{|z-1-i|=5/4} \frac{\operatorname{Log} z}{(z-1)^2} dz$

(e) $\oint_{|z|=1} \frac{dz}{z^2(z^2-4)e^z}$

(f) $\oint_{|z-1|=2} \frac{dz}{z(z^2-4)e^z}$

Solution

(a) The nominator has one zero at $z = 1$ inside $|z| = 2$, thus by by Cauchy´s theorem,

$$\oint_{|z|=2} \frac{z^n}{z-1} dz = \frac{2\pi i}{0!} [z^n]_{z=1} = 2\pi i.$$

(b) The nominator has no zeros inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{z^n}{z-2} dz = 0.$$

(c) The nominator has one zero at $z = 0$ inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{\sin z}{z} dz = \frac{2\pi i}{0!} [\sin z]_{z=0} = 0.$$

(d) Let f be the principal branch of the logarithm defined on the slit plane, that is, $f(z) = \operatorname{Log} z$ for all $z \neq 0$ with $\operatorname{Arg} z \neq -\pi$. Then f is analytic on a domain $|z - 1 - i| \leq 5/4$. The nominator has one double zero at $z = 1$ inside $|z - 1 - i| < 5/4$, thus by by Cauchy´s theorem.

$$\oint_{|z-1-i|=5/4} \frac{\operatorname{Log} z}{(z-1)^2} dz = \frac{2\pi i}{1!} \left[\frac{d}{dz} (\operatorname{Log} z) \right]_{z=1} = 2\pi i.$$

(e) The nominator has one double zero at $z = 0$ inside $|z| = 1$, thus by by Cauchy´s theorem,

$$\oint_{|z|=1} \frac{dz}{z^2 (z^2-4) e^z} = \frac{2\pi i}{1!} \left[\frac{d}{dz} \left(\frac{e^{-z}}{z^2-4} \right) \right]_{z=0} = \frac{\pi i}{2}.$$

(f) The nominator has one zero at $z = 0$, and one zero at $z = 2$ inside $|z - 1| = 2$, thus by by Cauchy´s theorem,

$$\oint_{|z-1|=2} \frac{dz}{z(z^2-4)e^z} = \frac{2\pi i}{0!} \left[\frac{1}{(z^2-4)e^z} \right]_{z=0} + \frac{2\pi i}{0!} \left[\frac{1}{z(z+2)e^z} \right]_{z=2} = -\frac{\pi i}{2} + \frac{\pi i}{4e^2}.$$